

Automated Localization and Classification of Thoracic Abnormalities in Chest Radiographs: AMIA Public Challenge 2026

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Abstract

This project addresses the AMIA Public Challenge 2026, which requires automated localization and classification of 14 thoracic abnormalities from chest radiographs — a multi-class object detection task with direct clinical relevance as a radiologist decision-support tool. The goal was to develop and compare detection models capable of jointly predicting bounding box locations and pathology labels across the dataset.

Four models were developed and evaluated: a UNet segmentation baseline (ResNet34 encoder), an anchor-free CNN detector (YOLOv8m), a transformer-based detector (RT-DETR), and a class-wise Weighted Box Fusion (WBF) ensemble. Annotation aggregation strategies (union, majority vote, WBF) were compared using UNet as a controlled baseline, with the union strategy adopted for all subsequent models. Confidence threshold tuning was applied to the best-performing single model, RT-DETR.

RT-DETR achieved the highest test performance (Kaggle mAP@IoU=0.4: 0.127), outperforming UNet (0.065), YOLOv8m (0.061), and the WBF ensemble (0.078). Despite weighting ensemble contributions by per-class validation mAP50, the ensemble underperformed RT-DETR alone, suggesting that validation-calibrated weights do not transfer reliably when constituent models generalize unevenly to the test set. Performance on rare classes (e.g., *Pneumothorax*) remained limited across all models, reflecting the fundamental challenge of class-imbalance in the datasets.

1 Introduction

Chest radiography is one of the most frequently performed medical imaging procedures, yet its interpretation remains challenging even for experienced radiologists. This project addresses the e AMIA Public Challenge ¹, requiring automated localization and classification of 14 distinct thoracic abnormalities—a combined multi-class object detection and classification task predicting bounding boxes and category labels per scan.

The dataset consists of 15,000 chest X-ray scans with bounding box annotations from three independent radiologists, presenting severe class imbalance and notable inter-radiologist spatial variation.

Four models were evaluated. **UNet**, an anchor-free CNN detector (**YOLOv8m**), and a transformer-based detector (**RT-DETR**). Finally, a **Weighted Box Fusion (WBF)** ensemble was applied as a post-processing step.

2 Exploratory Data Analysis

2.1 Class Distribution

Analysis of the training annotations show a severe imbalance across classes. As shown in Figure 1, Aortic enlargement is the most frequent finding with 5,481 annotations, whereas *Pneumothorax* appears only 195 times. "No finding", meaning it contains no abnormality had 15,525 out of 45,925 rows (33.8%), and at the image level, 5,175 out of 8,573 images (60.4%). This extreme imbalance renders standard baseline classifiers ineffective; models easily overfit to dominant background signals or frequent classes while failing to detect clinically critical rare pathologies. This structural imbalance justifies adopting dedicated object detectors (YOLOv8m and RT-DETR), which leverage explicit bounding-box spatial supervision.

¹<https://www.kaggle.com/competitions/amia-public-challenge-2026/overview>

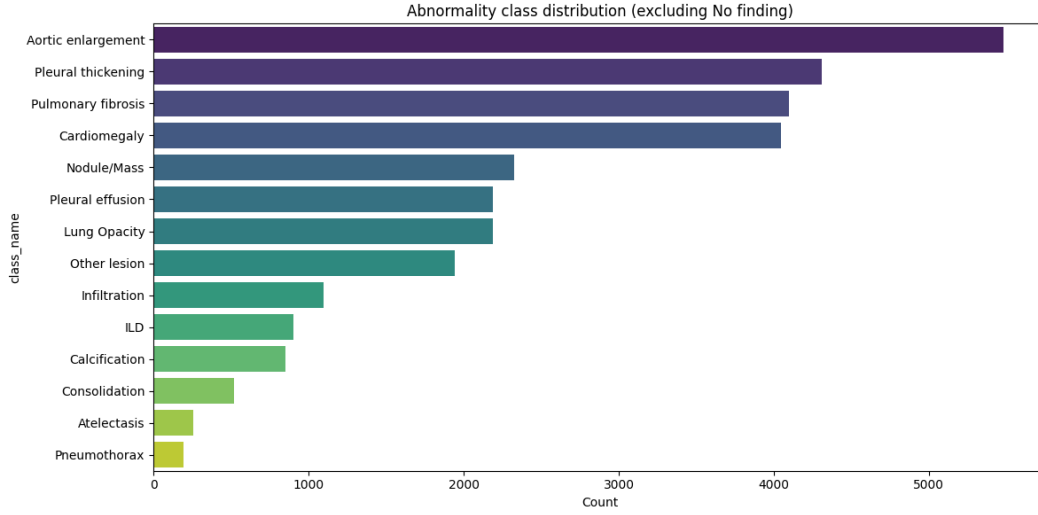


Figure 1: Class distribution excluding "No finding".

2.2 Multi-Radiologist Annotations

Annotations were contributed by 17 radiologists in total (R1–R17), and each image was reviewed by 3 radiologists selected from this pool. A key challenge from this setup is inter-radiologist disagreement. For the same image, different radiologists may identify different findings or draw bounding boxes at different locations.

Figure 2 shows an example of this.

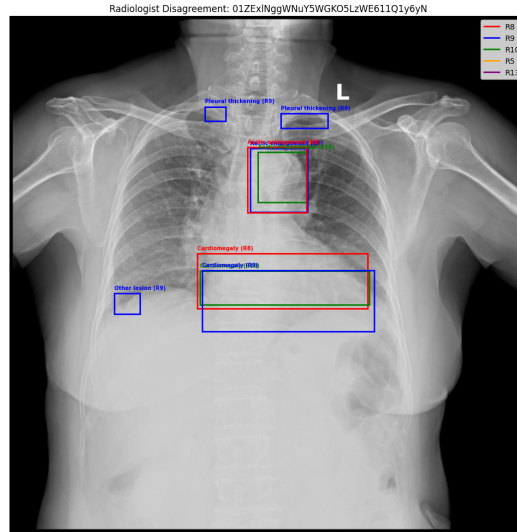


Figure 2: Example of inter-radiologist disagreement on a chest X-ray. Three radiologists (R8, R9, R10) unanimously identified Aortic enlargement and Cardiomegaly, but their bounding box placements differ noticeably.

2.3 Image Size and Coordinate Scaling

The original X-ray images have variable dimensions, with a mean size of $2,787 \times 2,446$ pixels. All images were pre-converted to 1024×1024 PNG format for the competition. However, bounding box coordinates in the annotation CSV are given in original image dimensions. Accordingly, all coordinates were rescaled to the 1024×1024 space prior to training.

2.4 Implications for Model Design

The data analysis reveals three key challenges that directly informed our model selection and training strategy. First, the severe class imbalance motivates the use of Varifocal Loss, which down-weights easy (background) examples and focuses training on hard, rare cases. Second, multi-radiologist disagreement requires a clear annotation aggregation strategy to avoid inconsistent supervision signals. Third, the co-occurrence of multiple findings per image necessitates a multi-label detection approach rather than single-class prediction.

3 Methods

3.1 Dataset description

The dataset for this competition consists of a total 15,000 postero-anterior (PA) CXR scans that have been annotated by experienced radiologists. From those, 8,573 images have been used for training and 6,427 were held-out for testing. The train csv contains 45,925 rows and the test csv 21,989.

All images are provided in PNG format, pre-converted from DICOM and resized to $1,024 \times 1,024$ pixels by the competition organizers. The challenge targets 14 thoracic abnormality classes—including *Aortic enlargement*, *Cardiomegaly*, *ILD*, *Nodule/Mass*, and *Pneumothorax*—along with a 15th category, "No finding" (class 14), indicating the absence of any abnormalities.

Every training image was reviewed by 3 radiologists, selected from a pool of 17 annotators (R1–R17), yielding 45,925 annotation rows in total. No images had missing annotations. While structurally complete, the dataset has inter-radiologist disagreement in both finding presence and bounding box placement, as detailed in the Section 2.1.

Additionally, annotation contribution was heavily skewed: three radiologists (R8, R9, R10) covered over 70% of all annotations (32,405 of 45,925). This imbalance raises the possibility of annotator-specific biases influencing the training distribution.

3.2 Experimental Protocol & Reproducibility

For the comparison across different models, key training conditions were standardized: 10 epochs, 512×512 input resolution, batch size of 8, and fixed random seed (42).

The 8,573 training scans were split into training (80%, 6,858 images) and validation (20%, 1,715 images) sets. Evaluation was conducted on the official held-out test set (6,427 images). Table 1 details the environment and hardware used for all experiments.

Table 1: System Configurations and Reproducibility Parameters

Component	Specification / Version
Python / CUDA	Python 3.12 / CUDA 12.8
PyTorch / Ultralytics	PyTorch 2.10.0+cu128 / Ultralytics 8.4.129
Libraries	<code>segmentation-models-pytorch</code> , <code>albumations</code> , <code>ensemble-boxes</code>
Hardware & Seed	NVIDIA Tesla T4 GPU (14GB VRAM) / Seed = 42

3.3 Pipeline Overview and Model Architectures

A four-step progressive workflow was established: UNet $\rightarrow$ YOLOv8m (anchor-free CNN) $\rightarrow$ RT-DETR (global transformer detector) $\rightarrow$ WBF Ensemble & RT-DETR Threshold Tuning.

UNet (Segmentation Baseline) utilizes a ResNet34 encoder pretrained on ImageNet with a standard decoder to output 14-channel spatial probability maps, trained using `BCEWithLogitsLoss` and Adam optimizer ($\text{lr} = 10^{-4}$). **YOLOv8m (Anchor-Free CNN Detector)**, pretrained on COCO, employs an anchor-free task-aligned assigner predicting bounding box centers and dimensions directly, trained with AdamW ($\text{lr}_0 = 10^{-3}$, cosine schedule). **RT-DETR (Transformer Detector)**, also pretrained on COCO, leverages self-attention for global context modeling and is trained using AdamW ($\text{lr}_0 = 10^{-4}$, Ultralytics default settings) with a composite loss combining GIoU, L1, and Varifocal classification losses.

3.4 Annotation Aggregation Strategy

Three strategies were evaluated using the UNet baseline: **Strategy 1 (Union)** retains all bounding boxes from all three radiologists as separate training targets to maximize supervision signals; **Strategy 2 (Majority Vote)** keeps only findings agreed upon by at least two radiologists; and **Strategy 3 (Weighted Box Fusion)** merges overlapping boxes ($\text{IoU} \geq 0.5$) into single consensus targets via weighted averaging. Given that Union best preserves the full spectrum of radiologist opinions present in the evaluation set, it was adopted as the standard target aggregation policy for all subsequent detection models (YOLOv8m and RT-DETR).

3.5 Preprocessing & Postprocessing

Preprocessing: Annotations were rescaled to 1024×1024 by the competition organisers, then resized to 512×512 to fit Kaggle GPU constraints (14GB VRAM). Images were converted from grayscale to

3-channel RGB via channel replication, normalized using ImageNet parameters ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$), and augmented with random horizontal flips ($p = 0.5$) and brightness/contrast adjustments ($p = 0.2$) using Albumentations.

Postprocessing & Rescaling:

- **UNet:** Probability masks of size 512×512 were thresholded at 0.3. Bounding boxes were extracted via connected component analysis (`scipy.ndimage.label`) and subsequently rescaled to the original 1024×1024 image dimensions (multiplied by factor 2.0).
- **YOLOv8 / RT-DETR:** Candidate bounding boxes predicted at 512×512 resolution were filtered using an optimized confidence threshold ($conf = 0.25$). All surviving box coordinates (x_1, y_1, x_2, y_2) were rescaled back to 1024×1024 by applying a spatial scaling factor of 2.0 ($1024/512$). Scans with no remaining candidate detections were assigned the competition placeholder 14 1.0 0 0 1 1 (Class 14: "No finding").

3.6 Ensemble via Weighted Box Fusion (WBF)

To leverage class-specific architectural strengths—such as RT-DETR’s superior modeling of global structural pathologies (e.g., Cardiomegaly) and YOLOv8’s high sensitivity to localized lesions (e.g., Nodules)—we fused candidate predictions from both models using Weighted Box Fusion (WBF). Unlike standard Non-Maximum Suppression (NMS), which aggressively discards overlapping candidate boxes and loses consensus information, WBF performs class-wise fusion by dynamically averaging predicted coordinates and confidence scores weighted by individual model reliability to reconcile inter-radiologist spatial variations.

4 Results

4.1 Design and Training Overview

A four-step progressive pipeline was developed: UNet $\rightarrow$ YOLOv8m $\rightarrow$ RT-DETR $\rightarrow$ WBF Ensemble.

- **UNet (ResNet34 Baseline):** Implemented first as a segmentation baseline due to its fast training convergence, enabling low-cost evaluation of multi-radiologist annotation aggregation strategies. Its core limitation—extracting bounding boxes post-hoc from segmentation masks without direct coordinate supervision—motivated transitioning to dedicated object detectors.
- **YOLOv8m (Anchor-Free CNN):** Selected as a localized CNN detector offering fast convergence and robust baseline localization. Beyond its individual performance, choosing YOLOv8m was strategically motivated by its shared ecosystem with RT-DETR within the Ultralytics framework.
- **RT-DETR:** Selected for its Vision Transformer backbone to model global receptive fields and self-attention context. This architecture provides a distinct advantage for detecting anatomically diffuse or structurally defined pathologies (e.g., *Cardiomegaly*, *ILD*) that local CNN filters struggle to capture. Furthermore, pairing RT-DETR with YOLOv8m established essential architectural diversity for model ensembling.
- **Weighted Box Fusion Ensemble:** Chosen over standard NMS to fuse predictions from RT-DETR and YOLOv8m using per-class validation mAP50 weights ($\text{IoU} = 0.5$). WBF constructs optimal consensus bounding boxes, directly accommodating inter-radiologist spatial variations and exploiting cross-architecture synergy.

To address dataset imbalance, we adopted a Union annotation strategy to preserve scarce positive signals and relied on Verifocal Loss within RT-DETR to suppress background noise. RT-DETR’s built-in composite loss (which natively employs Varifocal Loss for classification alongside GIoU and L1 losses) implicitly down-weighted easy background examples without requiring manual class-weight tuning.

4.2 Metrics

The task requires both identifying *what* finding is present and *where* it is located. Metrics were chosen to reflect both requirements.

Dice / IoU (UNet): Pixel-level overlap between predicted and ground truth masks, computed only on GT-positive images. Both metrics are ratio-based and scale-invariant; no rescaling from 512×512 to 1024×1024 is required. No pixel-unit metrics were used.

mAP50 (RT-DETR, YOLO): A predicted box is correct only if it overlaps the ground truth by $\geq 50\%$ IoU *and* the class label is correct. This jointly evaluates localization and classification. The competition’s Kaggle score uses mAP@IoU=0.4.

UNet’s Dice/IoU and detection mAP are not directly comparable. The Kaggle score (mAP@IoU=0.4 on held-out test set) serves as the common reference across all models.

4.3 Model Results

4.3.1 Summary

RT-DETR achieved the best test performance despite slightly lower validation mAP50 than YOLO, suggesting better generalization to unseen data. The ensemble result demonstrates that model combination does not always improve performance.

Table 2: Overall model comparison.

Model Architecture	Validation Metric	Public Score	Private Score
UNet	Mean Dice: 0.1315	0.06500	0.06500
YOLOv8m	Val mAP50: 0.233	0.06100	0.06100
Class-Weighted Ensemble (YOLOv8 + RT-DETR)	—	0.07800	0.07200
RT-DETR (Selected Optimal, $conf = 0.15$)	Val mAP50: 0.220	0.12700	0.12500
RT-DETR (Metric Peak, $conf = 0.25$)	Val mAP50: 0.220	0.12800	0.12600

4.3.2 UNet

Kaggle score: **0.065**. Most classes score near zero (Table 3) due to the model learning a “predict nothing” bias from 60.4% No-finding images. UNet demonstrates the fundamental limitation of segmentation models on detection tasks without explicit bounding box supervision.

Table 3: UNet per-class validation Dice and IoU (GT-only, selected classes).

Class	Dice	IoU
Cardiomegaly	0.4878	0.4119
Aortic enlargement	0.3676	0.3045
Pleural effusion	0.2297	0.1751
Pulmonary fibrosis	0.1779	0.1300
Most others	≈ 0.00	≈ 0.00
Mean	0.1315	0.1101

4.3.3 RT-DETR

Kaggle score: **0.127** (+95% over UNet). Highest performance on Aortic enlargement and Cardiomegaly — large, anatomically well-defined findings with abundant training data. Worst on *Pneumothorax* (195 annotations) and Atelectasis (255 annotations), consistent with their data scarcity. This directly reflects the class imbalance identified during EDA, where data scarcity severely restricted the model’s ability to learn robust feature representations for subtle, localized lesions. To mitigate this supervision bottleneck, the Union annotation strategy was adopted to retain all positive instances, while RT-DETR’s native Varifocal Loss implicitly suppressed background dominance during gradient updates. Overall recall (0.325) exceeded precision (0.269), reflecting the intentional deployment of a low confidence threshold (0.15) to minimize missed diagnoses in a clinical context.

Table 4: RT-DETR validation performance (selected representative classes).

Class Category / Name	Precision	Recall	mAP50	mAP50-95
<i>Top Performance</i>				
Aortic enlargement	0.440	0.661	0.481	0.320
Cardiomegaly	0.385	0.643	0.442	0.328
<i>Lowest Performance</i>				
Atelectasis	0.144	0.174	0.094	0.044
Pneumothorax	0.142	0.196	0.085	0.048
Mean (All 14 Classes)	0.269	0.325	0.220	0.119

4.3.4 YOLOv8m

YOLOv8m (Medium variant) was selected as an anchor-free CNN baseline to balance computational efficiency and feature representation capability under Kaggle GPU constraints.

Kaggle score: **0.061**. Validation mAP50 (0.233) exceeded RT-DETR (0.220) on some classes, but YOLO generalised poorly to the test set. *Pneumothorax* is a notable failure: precision=1.0 but recall=0.0.

Table 5: YOLOv8m per-class validation performance (selected classes).

Class	Precision	Recall	mAP50	mAP50-95
Aortic enlargement	0.596	0.614	0.633	0.375
Cardiomegaly	0.568	0.619	0.607	0.386
Pleural effusion	0.244	0.467	0.298	0.118
Pneumothorax	1.000	0.000	0.016	0.006
All (mean)	0.381	0.276	0.233	0.115

4.3.5 Class-wise Weighted Ensemble (RT-DETR + YOLO)

Per-class weights were computed proportional to each model’s validation mAP50, and predictions were fused using Weighted Box Fusion (IoU threshold: 0.5).

Kaggle score: **0.078**. Fusing YOLOv8m and RT-DETR via Weighted Box Fusion (WBF) yielded a Kaggle test score of 0.078, compared to 0.061 for YOLOv8m alone and 0.127 for RT-DETR alone.

4.4 What Was Optimized

4.4.1 Annotation Strategy Comparison

Three strategies for aggregating multi-radiologist annotations were evaluated using UNet as a controlled baseline (10 epochs, ResNet34 encoder, identical settings across strategies).

Table 6: UNet validation performance by annotation aggregation strategy (GT-only evaluation).

Strategy	Mean Dice	Mean IoU
All annotations (union)	0.1315	0.1101
WBF	0.1259	0.1039
Majority vote	0.0962	0.0788

The Union strategy outperformed both alternatives and was adopted for all subsequent training. Beyond its superior validation score (Mean Dice: 0.1315, Table 6), its selection was driven by the class imbalance inherent in the dataset. In long-tailed pathology distributions (e.g., *Pneumothorax* with only 195 total instances), consensus-based filtering such as Majority Voting prematurely discards critical single-annotator positive signals. The Union strategy mitigates this data scarcity bottleneck by preserving all candidate annotations and maximizing available supervision for rare classes.

4.4.2 Confidence Threshold Analysis

While model loss functions and optimizers were maintained during training, RT-DETR was selected for further optimization as it showed the highest single-model performance on the benchmark. To refine its predictions, a systematic grid search over the confidence threshold ($conf \in [0.05, 0.25]$). Because the competition mAP metric penalizes false positives, this threshold calibration was essential to eliminate low-confidence noise on normal radiographs and to maximize precision.

This optimization addressed the precision-recall trade-off: $conf=0.25$ yields the best score (0.128), though the gain over baseline is marginal (+0.001). At $conf=0.05$, excessive false positives (215,614 boxes; 33.5 per image) severely degrade precision.

Table 7: RT-DETR prediction metrics and Kaggle test benchmark performance across confidence thresholds.

Conf. Threshold	Total Pathologies	Avg Boxes/Image	No Finding (%)	Public Score	Private Score
0.05	215,614	33.55	53.5%	0.11300	0.11200
0.10	77,018	11.98	68.3%	0.12300	0.12200
0.15 (Primary)	38,034	5.92	73.8%	0.12700	0.12500
0.20	22,902	3.56	76.8%	0.12700	0.12500
0.25	15,590	2.43	79.2%	0.12800	0.12600

While $conf = 0.25$ achieved the highest Public Kaggle Score (0.12800 vs. 0.12700), it produced a "No finding" ratio of 79.2%, significantly over-estimating normal scans compared to the EDA target (60.4%, see Figure S1). In a clinical diagnostic workflow, this over-filtering risks False Negatives. Because the gain from $conf = 0.25$ is negligible (+0.001), $conf = 0.15$ was chosen as our optimal primary threshold, balancing False Positive suppression with clinical diagnostic sensitivity.

5 Discussion

5.1 Reflections on Results

Annotation Agreements. Counterintuitively, the Union strategy outperformed "cleaner" methods (WBF, Majority Vote). In our imbalanced dataset, majority voting disproportionately discards rare-class annotations (e.g., *Pneumothorax*), and WBF reduces the overall box count. The Union strategy prevents this by maximizing the scarce positive supervision signal.

What Went Right: Transitioning from segmentation-to-box conversion (UNet) to direct anchor-free detection (RT-DETR) yielded the single largest performance improvement. Retaining multi-radiologist union annotations prevented label suppression.

Challenges:

Confident Thresholds. $conf = 0.25$ yields the best score (0.128), though the gain over baseline ($conf = 0.15$) is marginal (+0.001). However, No Finding ratio at $conf=0.25$ (79.2%) exceeds the ground truth (60.4%), indicating over-prediction of the negative class — yet this still maximises mAP, reflecting the metric’s sensitivity to precision over recall. Consequently, striking an optimal balance between maximizing mAP precision and maintaining clinical diagnostic recall remains challenging, when the "No Finding" ratio is too high in the dataset.

Ensemble Model. The WBF ensemble unexpectedly underperformed compared to the single RT-DETR model (0.07800 vs. 0.12700). YOLO’s weak test-set generalisation might have (0.06100) degraded RT-DETR’s higher-confidence predictions, demonstrating that validation-based ensemble weights do not always transfer to the test distribution.

5.2 Comparison to Winning Solutions

Top-ranking competition solutions (mAP 0.52–0.55) relied on 10-fold cross-validation, high-resolution inputs (640×640), and multi-model ensembling with confidence-based reranking across detector families (RT-DETR, DINO) and optimizer strategies (AdamW, MuSGD). The 3rd-place solution² (score: 0.532) achieved its final result by treating cross-validation folds as complementary detectors and using MuSGD predictions to rerank AdamW anchor confidences, rather than naive box fusion. In contrast, our approach trained a single RT-DETR model for 10 epochs at 512×512 resolution, achieving a Kaggle score of 0.127.

5.3 Future Work

- **Adaptive Ensemble Weighting:** Optimizing fusion weights based on bounding box scales or prediction uncertainty could better resolve inter-radiologist coordinate discrepancies.
- **Two-Stage Cascaded Architecture:** Given that "No Finding" scans constitute over 60% of the dataset, deploying a high-precision binary classifier as a first-stage filter could eliminate true negatives early. This would reduce unnecessary background predictions before routing candidate scans to the RT-DETR object detector.
- **Resolution & Training Scale-Up:** Fundamental performance improvements will require scaling training resolution (e.g., 1024×1024) to preserve subtle micro-lesions, alongside extended training schedules and stronger domain-specific augmentations.
- **Increasing training epochs.**

6 Code

All code pipelines and Kaggle notebooks are accessible at:

<https://github.com/hj-Hyunji/AMIA-Public-Challenge-2026>.

²https://vsokhoyan.github.io/ChestXRay_Multiclass_Detection/

7 Supplementary Figures

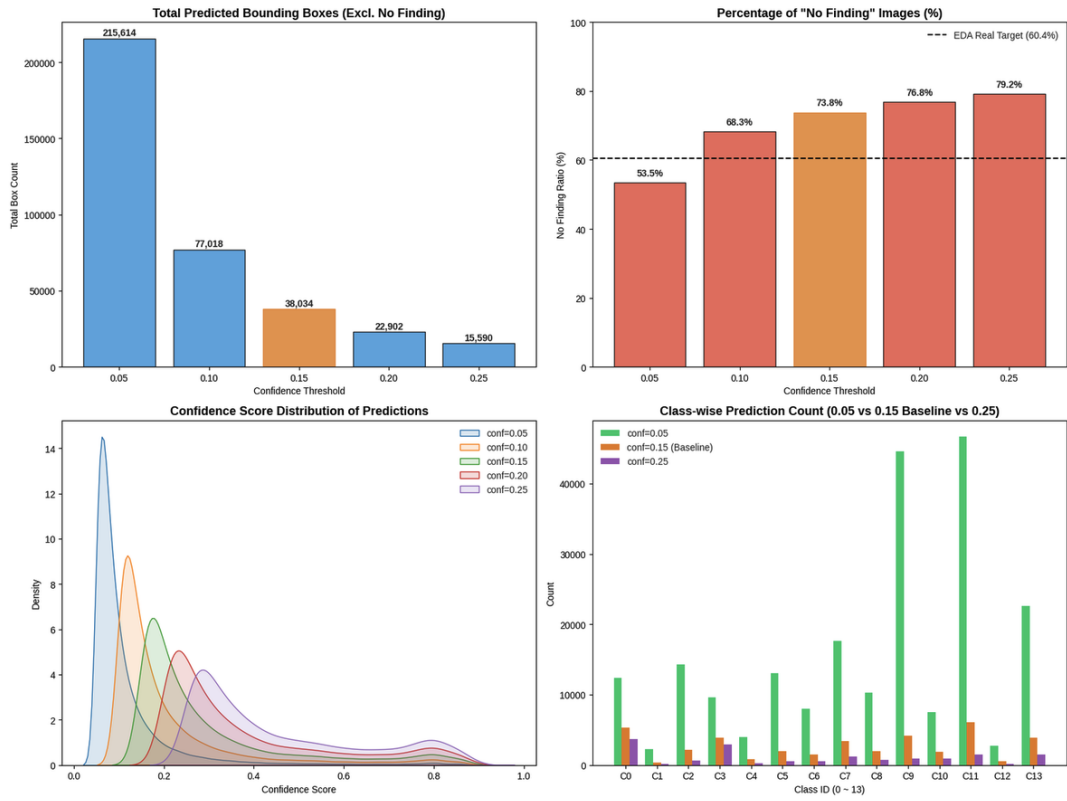


Figure S1: Analysis of RT-DETR prediction metrics across varying confidence thresholds ($\text{conf} \in [0.05, 0.25]$). **Top-left:** Total predicted bounding boxes dropping sharply as threshold increases. **Top-right:** Percentage of predicted “No Finding” scans compared against the ground-truth target (60.4%). **Bottom-left:** Confidence score density distribution curves. **Bottom-right:** Class-wise prediction count comparisons.